

Meta Ad Performance Analysis — Project Report

Facebook vs. Instagram campaign performance, audience targeting accuracy, and budget utilization- built on a 4-table star schema, 400,000 ad events, and 19 DAX measures.

Executive Summary :-

This report analyzes Meta (Facebook + Instagram) advertising performance across 200 ads, 50 campaigns, and 9,841 users, using a star-schema Power BI dashboard. Every figure in this report was independently recomputed from the underlying data and cross-checked against the dashboard's own DAX measures - not copied from the project's earlier insights document. That re-computation surfaced one central finding: the audience these ads are targeted at and the audience that actually engages are close to opposites. Ads are targeted at women 43% of the time, but men generate the majority (55%) of real engagement. Three smaller claims from the earlier insights document - an evening engagement peak, a specific top-5 country ranking, and an implicit combined-platform view - did not hold up under verification and have been corrected below.

1. About This Dataset

The dataset is modelled after how Meta's ad platforms (Facebook/Instagram) capture performance data, structured as a star schema:

- `ad_events` (fact table, 400,000 rows) - every impression, click, share, comment, and purchase, timestamped and tagged with day-of-week / time-of-day
- `ads` (dimension, 200 rows) - platform, creative type, and targeting per ad
- `campaigns` (dimension, 50 rows) - budget and timeframe per campaign
- `users` (dimension, 9,841 rows) - demographics and interests per user

Two things worth flagging about the source data itself: `event_type` actually contains six values in the raw data (Impression, Click, Share, Comment, Purchase, Like), not the five documented in the original schema notes - Like (12,013 events, 3.0% of the total) isn't part of any measure in the dashboard. And `ad_platform` only ever takes two real values, Facebook and Instagram - Messenger and Audience Network, mentioned as possible platforms, don't appear in this data at all.

2. Business Objective & Scope

The business need was a performance-tracking report for Facebook and Instagram ad campaigns, giving the marketing team visibility to identify the stronger platform, track ROI and budget allocation, and understand audience engagement - paid ads only, on Facebook and Instagram only.

3. KPI Definitions

KPI	Formula	What it tells you
Impressions	Count of event_type = Impression	Reach
Clicks	Count of event_type = Click	Engagement intent

Engagements	Clicks + Shares + Comments	Overall interaction volume
CTR	Clicks / Impressions	Ad effectiveness
Engagement Rate	Engagements / Impressions	Overall ad appeal
Conversion Rate	Purchases / Clicks	Funnel efficiency once someone's already interested
Purchase Rate	Purchases / Impressions	Overall odds a view becomes a sale
Total Budget	Sum of campaign budgets	Cost basis

4. Dashboard Insights

The earlier insights document reported one set of KPIs (216K impressions, 25.4K clicks, 11.76% CTR) without stating which platform they came from. Recomputing independently and checking both platforms shows those numbers are Facebook only - Instagram runs at roughly half Facebook's volume, with slightly different rates:

Metric	Facebook	Instagram
Impressions	215,972	123,840
Clicks	25,389	14,690
Purchases	1,323	708
Engagements	29,296	16,848
CTR	11.76%	11.86%
Engagement Rate	13.56%	13.60%
Conversion Rate	5.21%	4.82%
Purchase Rate	0.61%	0.57%

Total Budget across all 50 campaigns (not platform-specific - budget is set at the campaign level, and a campaign can run ads on both platforms): \$2,535,924, averaging \$50,718 per campaign.

Reading the funnel: both platforms show the same shape - strong top-of-funnel performance (CTR comfortably above the roughly 1-2% industry benchmark for paid social) but a steep drop from click to purchase. Facebook converts that interest into purchases slightly better than Instagram (5.21% vs. 4.82%), which combined with its 2× volume likely means Facebook drives meaningfully more total purchases even though the two platforms look similar on engagement alone.

5. Key Findings

5.1 -Targeting and reality point in different directions

Breaking Facebook engagement down by who the ad was targeted at (`ads.target_gender`) shows a female-leaning split: Female 43%, "All" 35%, Male 22%. Breaking the exact same engagements down by the user's actual recorded gender (`users.user_gender`) shows something close to the opposite: Male 55%, Female 35%, Other 10%. Whatever is driving male users to engage, it isn't the stated targeting strategy - this is the single most actionable finding in the data, because it directly answers the BRD's own stated goal of checking whether targeting matches actual engagement.

5.2 - An "evening peak" that isn't there

The earlier insights document described engagement as peaking "around late afternoon & evening (15-20 hours)" with the lowest engagement in early morning. Recomputed directly from the `time_of_day` field, Facebook engagement is essentially flat: Morning 7,309 · Afternoon 7,330 · Evening 7,382 · Night 7,275 - a spread of under 1.5% between the highest and lowest bucket. There's no peak to schedule around in this data. (The separate claim that engagement is consistent across the days of the week does hold up - 4,128 to 4,274 across all seven days, also essentially flat, which is what the original document said.)

5.3 - The top-countries list doesn't reproduce

The earlier document named "US, India, Brazil, Germany, UK" as the top engaged countries. Ranking every country by actual event volume gives a different order: United States, United Kingdom, Canada, India, Germany - with Brazil down in 7th place and Australia (not mentioned at all in the original list) in 6th.

5.4 - Best-performing ad format

Across all four creative formats, Stories has both the highest volume (72K impressions) and the highest Purchase Rate (0.65%); Video has the highest CTR (11.9%) and Engagement Rate (13.7%). Carousel and Image trail on every metric - consistent with the earlier document's recommendation to prioritize Video and Stories, which does hold up under recomputation.

5.5 - An undocumented, uncounted event type

Like events (12,013 total, split roughly 7,505 Facebook / 4,508 Instagram) exist in the raw data but appear in neither the original schema documentation nor the Engagements DAX measure. This isn't necessarily wrong - a like is a much lower-effort action than a comment or share - but it's an undocumented gap rather than a deliberate, stated design choice, and worth an explicit decision either way.

6. Final Recommendations

Investigate the targeting mismatch before spending more on it. Either the targeting strategy needs to shift toward the audience that's actually responding, or the female-targeted spend is reaching a real but different audience than intended - both are worth knowing, and neither is visible without this comparison.

Shift budget toward Stories and Video, which outperform Carousel and Image on every metric measured.

Drop the evening-scheduling recommendation. It isn't supported once the underlying counts are checked - reallocating effort based on it wouldn't be expected to move engagement.

Resolve the Like-event question explicitly - include it in Engagements, or document why it's deliberately excluded.

Update country-targeting priorities to reflect the verified ranking (US, UK, Canada, India, Germany) rather than the original list.

7. Methodology Note

This project's source Raw Data folder was not available at analysis time; all figures in this report were computed from the dataset recovered directly from the Power BI file's embedded data model, then verified by independently

reproducing every KPI in the project's original insights document. Full methodology and a complete list of checks:
logs/data_processing_log.md in the project repository.